ILKIN ISLERan
ilkinisler@gmail.com | www.linkedin.com/in/ilkinsevgiisler

PROFESSIONAL SUMMARY

Ph.D. in Computer Science with deep expertise in AI and deep learning. Experienced in designing and deploying robust, explainable AI systems for high-risk applications. Skilled in computer vision, uncertainty quantification, model interpretability, and advanced NLP workflows including retrieval-augmented generation (RAG), citation systems, hallucination detection, and large language models (LLMs). Highly adaptable and passionate about continuous learning, with a strong drive to apply AI and machine learning to solve complex challenges and create meaningful impact.

EDUCATION

- **Ph.D.**, Computer Science, **University of Central Florida**, Orlando, FL — Aug 2025
- **MS**, Computer Science, **University of Central Florida**, Orlando, FL — Dec 2022
- **BS**, Computer Science, **Hacettepe University, Ankara**, Turkey — June 2020

WORK EXPERIENCE

Senior AI Engineer — Aug 2025 - Present
Universal Creative, AI Strategy & Solutions — *Orlando, FL*
- Architected and built a secure, production-grade MCP-based AI platform from scratch, enabling centralized orchestration and lifecycle management of AI agents and tools. Designed and implemented role-based access control (RBAC), authentication layers, and security hardening mechanisms to ensure safe deployment of ML services at scale. Delivered an end-to-end infrastructure stack supporting data ingestion, model integration, API interfaces, and governed execution of AI workflows.

AI Software Engineer — Jan – Aug 2025
Universal Creative, Engineering & Safety — *Orlando, FL*
- Developed AI solutions using LLMs and retrieval-augmented generation (RAG) with capabilities for hallucination detection, groundedness evaluation, citation, meta tagging and topic modeling.

Graduate Teaching Assistant — Aug 2023 – Dec 2024
University of Central Florida, College of Engineering and Computer Science — *Orlando, FL*
- Assisting with Advanced Artificial Intelligence and Advanced Computer Networks courses.

Graduate Research Assistant — Jan 2020 – Aug 2023
University of Central Florida, College of Engineering and Computer Science — *Orlando, FL*
- Built explainable, uncertainty-aware models for organ and tumor segmentation and outcome prediction.
- Led cross-institutional projects (including Northwestern Radiology) to deploy AI visualization tools.
- Oversaw project delivery and documentation to support clinical integration of AI models.

Python Developer — May - Oct 2022
Universal Orlando Resort, BAIE Department — *Orlando, FL*
- Enhanced system health and stability by identifying and resolving 100+ vulnerabilities in cloud scripts.
- Optimized machine learning pipelines with Azure Databricks and MLflow to enhance performance.

Data Analyst — Jan - Aug 2020
Simsoft Information Technologies — *Ankara, Turkey*
- Implemented LSTM models for predictive maintenance and anomaly detection.
- Simulated COVID-19 spread using models to analyze disease transmission and population impact.

Research Intern — Jul - Sep 2019
University of Central Florida, Synthetic Reality Lab — *Orlando, FL*
- Utilized Generative Adversarial Networks (GANs) for facial expression recognition.

Data Analyst Intern — Jul - Sep 2018
Carbon Health — *San Francisco, CA*
- Utilized machine learning to predict outcomes, validate data and reduce clinic wait times by 30%, improving appointment forecasting and overall efficiency.

TECHNICAL SKILLS

Languages: Python, Java, C, SQL
ML Systems & Infrastructure: MCP-based AI systems architecture, distributed ML pipelines, model serving APIs, RBAC, secure AI deployment, model governance,
Areas of Expertise: Computer Vision; NLP (LLMs, RAG, hallucination detection); Uncertainty Quantification; Explainable AI; Anomaly Detection; ML Systems Design

ACADEMIA

I. Journals

- Zhang, Z. et al., *"Large-scale multi-center CT and MRI segmentation of pancreas with deep learning."* Medical Image Analysis, Volume 99, January 2025, 103382. doi:10.1016/j.media.2024.103382. (co-author)
- Gulec, U.*, **Isler, I. S.*,** et al., *"Power-VR: Interactive 3D virtual environment to increase motivation levels of powerlifters during training sessions."* Computer Animation and Virtual Worlds, Volume 34, Issue 2, 2022, e2045. doi: 10.1002/cav.2045. (*equal contribution)

II.Peer- reviewed conference articles (full papers)

- **Isler, I. S.** et al., *" Uncertainty-Guided Coarse-to-Fine Tumor Segmentation with Anatomy-Aware Post-Processing."* **ACDSA IEEE, 2025**. doi: 10.48550/arXiv.2504.12215.
- **Isler, I. S.** et al., *"Self-Supervised Learning for Organs At Risk (OARs) and Tumor Segmentation with Uncertainty Quantification."* **ICECCME IEEE, 2023**. doi: 10.1109/ICECCME57830.2023.10252269
- **Isler, I. S.** et al., *"Enhancing Organ at Risk Segmentation with Improved Deep Neural Networks."* **SPIE Medical Imaging, 2022**. doi: 10.1117/12.2611498.
- Ali, K. et al., *"Facial Expression Recognition Using Human to Animated-Character Expression Translation."* arXiv preprint, 2019. doi:10.48550/arXiv.1910.05595. (co-author)

III.Other publications

- Jha, D. et al., *"Practical and Ethical Considerations for Generative AI in Medical Imaging."* **FAIMI & EPIMI at MICCAI 2024**, pp. 176–187. doi:10.1007/978-3-031-72787-0_17. (co-author)
- **Isler, S.** et al., *"Usage of 3D Modelling and Morphing Tools in Plastic Surgery."*, **UYMS**, **2019**.

IV.Peer Reviewing

- Medical Physics, American Association of Physicists in Medicine
- IEEE, Journal of Biomedical and Health Informatics
- IEEE, Transactions on Medical Imaging

ADDITIONAL CREDENTIALS

Workshops & Conferences

- Presenter, *SPIE Medical Imaging 2022,* San Diego, CA — Feb 2022
- Participant, *MIDL Doctoral Symposium,* Virtual — July 2021
- Participant, *Trustworthy AI for Imaging-based Diagnoses (IEEE EMBS),* Virtual — June 2021
- Team member, *UCF Research Experiences (NSF REU),* Orlando, FL — May 2021

Honors & Awards

- Fellowships
  - MI22 Student Conference Support — Dec 2021
  - UCF College of Graduate Studies Presentation Fellowship — Nov 2021
- Competitive powerlifter
  - 4 x USA Powerlifting FL state record holder
  - 2 x Powerlifting Overall Champion of Turkey
  - IPSU European Championship – 2nd Place Overall